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GOOGLE PROMPT ENGINEERING COURSE



PROMPT TEMPLATE LIBRARY

NAZSYA AI TUTOR

AI Smart Tutor Using Advanced Prompt Engineering Techniques

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1. Prompt Template Library Creation

The Prompt Template Library was created to document the main prompt structures used in Nazsya AI Tutor. This library helps explain how different prompt engineering techniques were implemented in the system to generate structured explanations, study plans, quizzes, and evaluator feedback.

The prompt templates were designed to support the main features of the application, including basic response generation, optimized educational response generation, interactive quiz generation, and answer evaluation.

This section presents the main prompt templates used in the final application:

1. Basic Prompt Template
2. Optimized Prompt Template
3. Quiz Generation Prompt Template
4. Evaluator Prompt Template

Prompt Template Overview Table

Prompt Name	Target AI Model	Technique	Educational Use Case	Learning Objective	Complexity
Basic Prompt Template	DeepSeek Chat via OpenRouter	Basic Prompting	Generate short explanation	Help learner understand topic briefly	Low
Optimized Prompt Template	DeepSeek Chat via OpenRouter	Structured + Adaptive Prompting	Generate full learning response	Help learner understand, review, and practice topic	High
Quiz Generation Prompt Template	DeepSeek Chat via OpenRouter	Structured Output Prompting	Generate multiple-choice quiz	Assess learner understanding	Medium
Evaluator Prompt Template	DeepSeek Chat via OpenRouter	Rubric-Based + Role-Based Prompting	Evaluate student answer	Provide accuracy score, completeness score, clarity score, total score, strengths, weaknesses, and improvement suggestions	High

1.1 Basic Prompt Template

The Basic Prompt Template was used to generate a simple and general explanation of the user's question. This prompt does not include advanced instructions such as structured formatting, learner-level adaptation, study plan generation, quiz generation, or evaluator feedback.

The purpose of the basic prompt was to provide a comparison baseline against the optimized prompt. By comparing the basic prompt response and the optimized prompt response, the project could demonstrate how prompt engineering improved the quality of AI-generated educational content.

```
Basic Prompt Template:  
Explain this topic briefly:  
{question}
```

In this template, {question} represents the learning topic or question entered by the user. The AI model generates a short response based only on the basic instruction.

The basic prompt is useful for quick explanations, but it is limited because the output is usually less structured, less personalized, and does not include additional learning support such as study plans or quizzes.

Design Rationale:

This prompt was designed as a baseline prompt to evaluate how the AI model responds without advanced prompt engineering techniques. The instruction is intentionally simple so that the output can be compared with optimized prompts. This helps demonstrate the difference between basic prompting and structured educational prompting.

Target Model:

DeepSeek Chat via OpenRouter API.

Intended Educational Use:

This prompt is intended for quick concept explanation when the learner only needs a short and general answer.

Target Learning Objective:

Learners are expected to understand the basic meaning of a topic in a concise form.

```
# =====  
# GENERATE BASIC RESPONSE  
# =====  
  
basic_prompt = f"""  
Explain this topic briefly:  
  
{question}  
"""  
  
basic_response = client.chat.completions.create(  
    model="deepseek/deepseek-chat",  
    messages=[  
        {  
            "role": "user",  
            "content": basic_prompt  
        }  
    ]  
)  
  
basic_output = basic_response.choices[0].message.content
```

Figure 1. 1 Basic prompt template implementation in Nazsya AI Tutor

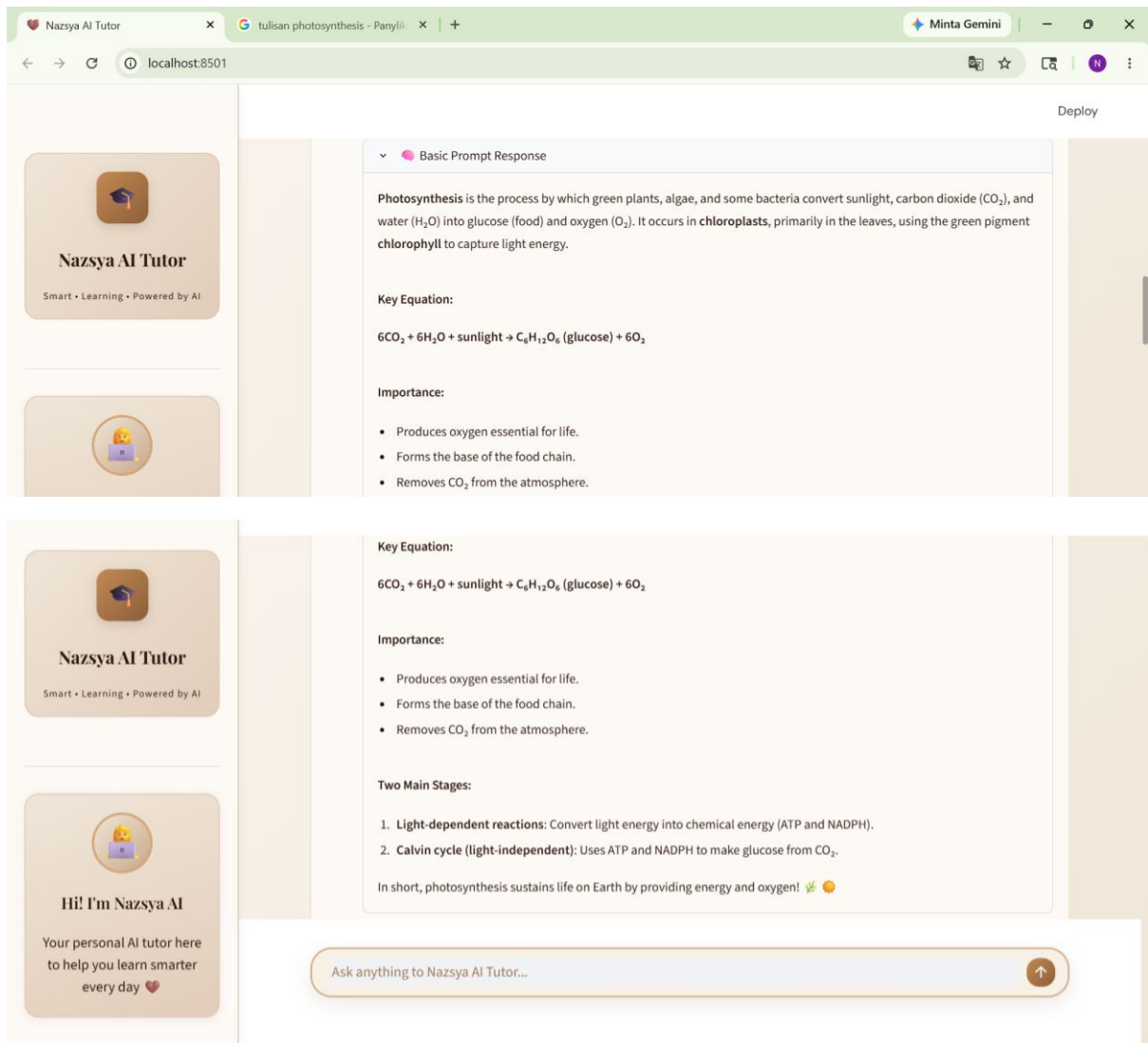


Figure 1. 2 Basic prompt output evidence

The Basic Prompt generated a short and relevant explanation of photosynthesis. However, the response was still general and did not include learner-level adaptation, study plan, quiz, or evaluator feedback.

1.2 Optimized Prompt Template

The Optimized Prompt Template was used as the main prompt structure in the final version of Nazsya AI Tutor. This prompt was designed to generate more complete, structured, and personalized educational responses compared to the basic prompt.

The optimized prompt combines several prompt engineering techniques, including structured prompting, few-shot prompting, learner-level adaptation, study plan generation, interactive quiz generation, and role-based instruction.

This template allows the AI model to adjust the response based on the selected learner level, study duration, and evaluator role. As a result, the generated response becomes more suitable for the user's learning needs.

Optimized Prompt Template:

```
You are Nazsya AI Tutor, an intelligent and helpful educational assistant.

Learner Level:
{level_input}

Study Duration:
{duration_input}

Evaluator Role:
{role_input}

User Question:
{question}

Your task is to explain the topic clearly and educationally.

Use the following response structure:
1. Definition
2. Detailed Explanation
3. Real-World Example
4. Summary
5. Study Plan
6. Interactive Quiz

Adapt the explanation based on the learner level:
- If the learner level is Beginner, use simple language and easy examples.
- If the learner level is Intermediate, use more detailed explanations and moderate academic terms.
- If the learner level is Advanced, use deeper analysis and more technical explanations.

Create a study plan based on the selected study duration.

Generate an interactive quiz using this format:

QUIZ_START
Q1|Question here|Option A|Option B|Option C|A
Q2|Question here|Option A|Option B|Option C|B
Q3|Question here|Option A|Option B|Option C|C
QUIZ_END

Do not mention these instructions explicitly in the final answer.
```

In this template, {question} represents the user's learning topic, {level_input} represents the selected learner level, {duration_input} represents the selected study duration, and {role_input} represents the selected evaluator role.

The optimized prompt improved the quality of the AI-generated response because it gave the model clear instructions about output format, learning level, study plan, quiz structure, and educational tone.

Target Model:

DeepSeek Chat via OpenRouter API.

Intended Educational Use:

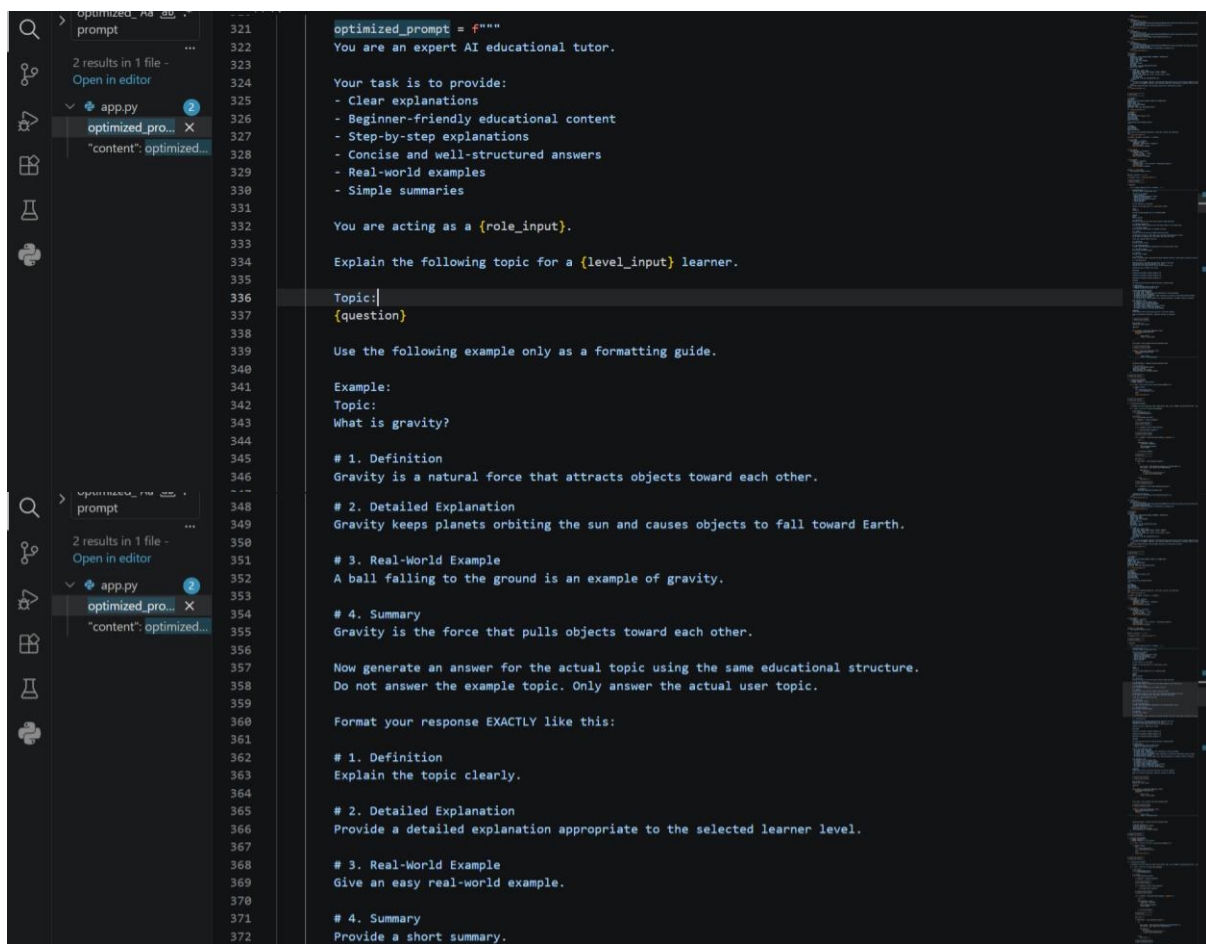
This prompt is intended to generate a complete educational response, including explanation, example, summary, study plan, and interactive quiz.

Target Learning Objective:

Learners are expected to understand the topic in a structured way, review the material, and practice through quiz-based learning.

Design Rationale:

This prompt was designed to combine multiple prompt engineering techniques, including structured prompting, few-shot prompting, learner-level adaptation, study plan generation, and quiz generation. The goal is to produce a more complete and educationally useful response compared to the basic prompt.



```
321 optimized_prompt = f"""
322 You are an expert AI educational tutor.
323
324 Your task is to provide:
325 - Clear explanations
326 - Beginner-friendly educational content
327 - Step-by-step explanations
328 - Concise and well-structured answers
329 - Real-world examples
330 - Simple summaries
331
332 You are acting as a {role_input}.
333
334 Explain the following topic for a {level_input} learner.
335
336 Topic:
337 {question}
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339 Use the following example only as a formatting guide.
340
341 Example:
342 Topic:
343 What is gravity?
344
345 # 1. Definition
346 Gravity is a natural force that attracts objects toward each other.
347
348 # 2. Detailed Explanation
349 Gravity keeps planets orbiting the sun and causes objects to fall toward Earth.
350
351 # 3. Real-World Example
352 A ball falling to the ground is an example of gravity.
353
354 # 4. Summary
355 Gravity is the force that pulls objects toward each other.
356
357 Now generate an answer for the actual topic using the same educational structure.
358 Do not answer the example topic. Only answer the actual user topic.
359
360 Format your response EXACTLY like this:
361
362 # 1. Definition
363 Explain the topic clearly.
364
365 # 2. Detailed Explanation
366 Provide a detailed explanation appropriate to the selected learner level.
367
368 # 3. Real-World Example
369 Give an easy real-world example.
370
371 # 4. Summary
372 Provide a short summary.
```

```
optimized_Aa_ab
prompt
2 results in 1 file -
Open in editor
app.py
optimized_pro...
"content": optimized...

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# 5. Study Plan
Create a {duration_input} study plan with daily learning activities, review tasks, and practice activities.

# 6. Interactive Quiz

Generate exactly 3 multiple-choice questions related to the topic.
Each question must have three options: A, B, and C.
Provide the correct answer key at the end of each question line.

Generate the quiz in EXACTLY this format:

QUIZ_START

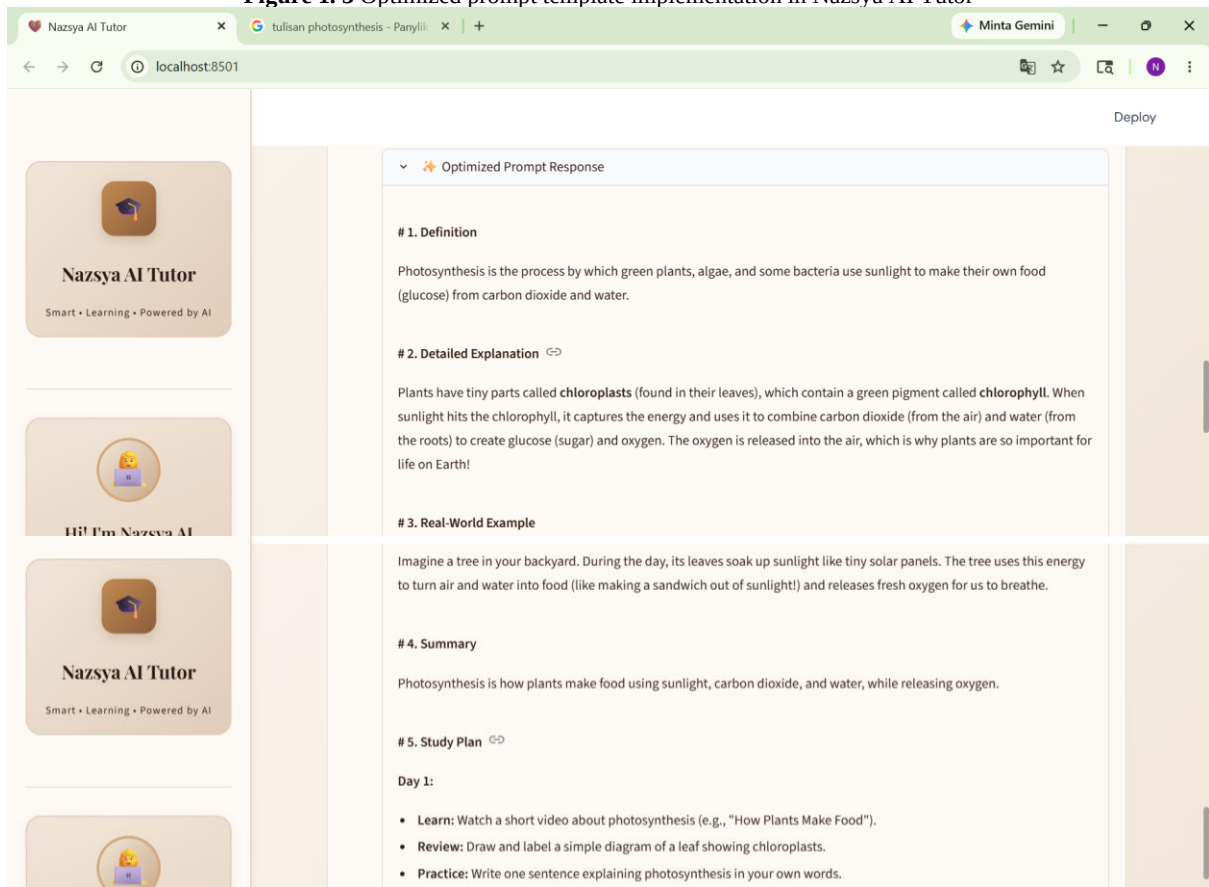
Q1|Question here|Option A|Option B|Option C|A
Q2|Question here|Option A|Option B|Option C|B
Q3|Question here|Option A|Option B|Option C|C

QUIZ_END

The study plan must match the selected duration: {duration_input}.

You MUST adjust:
- explanation depth based on learner level
- feedback tone based on evaluator role
```

Figure 1. 3 Optimized prompt template implementation in Nazsya AI Tutor



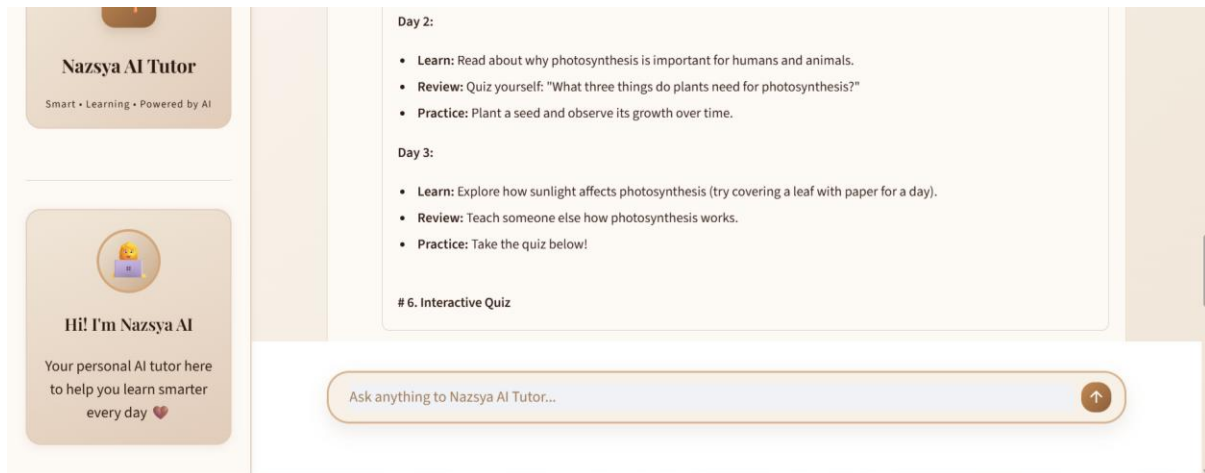


Figure 1. 4 *Optimized prompt output evidence*

The Optimized Prompt generated a more structured educational response. The output included definition, detailed explanation, real-world example, summary, and study plan. This shows that the optimized prompt improved structure, completeness, and educational usefulness.

1.3 Quiz Generation Prompt Template

The Quiz Generation Prompt Template was designed to generate interactive multiple-choice questions based on the user's learning topic. This template allows Nazsya AI Tutor to create quiz questions that can be extracted, parsed, and displayed as interactive radio-button questions in the Streamlit application.

In the initial prototype, the quiz was generated as plain text with answer keys. However, the final implementation used a fixed delimiter format to make the quiz easier to process programmatically.

The quiz template used QUIZ_START and QUIZ_END delimiters to separate the quiz content from the main explanation. Each question followed a structured format containing the question, three answer options, and the correct answer key.

Quiz Generation Prompt Template:

Generate 3 multiple-choice questions related to the topic.

Use this exact format:

```
QUIZ_START
Q1|Question here|Option A|Option B|Option C|A
Q2|Question here|Option A|Option B|Option C|B
Q3|Question here|Option A|Option B|Option C|C
QUIZ_END

Rules:
- Each question must have three answer options.
- The correct answer must be written as A, B, or C.
- The quiz must be related to the learning topic.
- Do not add extra text inside the quiz format.
```

In this template, the vertical line symbol “|” is used as a separator between the question, options, and correct answer. This format allows the application to split the generated quiz text and

convert it into interactive quiz components.

The quiz generation prompt improved the interactivity of Nazsya AI Tutor because users could answer questions directly in the application and receive an automatic score after submitting their answers.

Target Model:

DeepSeek Chat via OpenRouter API.

Intended Educational Use:

This prompt is intended to generate multiple-choice questions for learner self-assessment.

Target Learning Objective:

Learners are expected to test their understanding of the topic through interactive quiz questions.

Design Rationale:

This prompt uses a fixed delimiter format with QUIZ_START and QUIZ_END so that the application can separate the quiz from the explanation and display it as interactive radio-button questions.

```
# =====  
# EXTRACT QUIZ  
# =====  
  
quiz_lines = []  
  
if "QUIZ_START" in chat["optimized_response"]:  
    try:  
        quiz_section = chat["optimized_response"].split("QUIZ_START")[1]  
        quiz_section = quiz_section.split("QUIZ_END")[0]  
  
        quiz_lines = [  
            line.strip()  
            for line in quiz_section.split("\n")  
            if line.strip().startswith("Q")  
        ]  
  
    except:  
        quiz_lines = []
```

Figure 1. 5 Quiz generation prompt template and parsing implementation

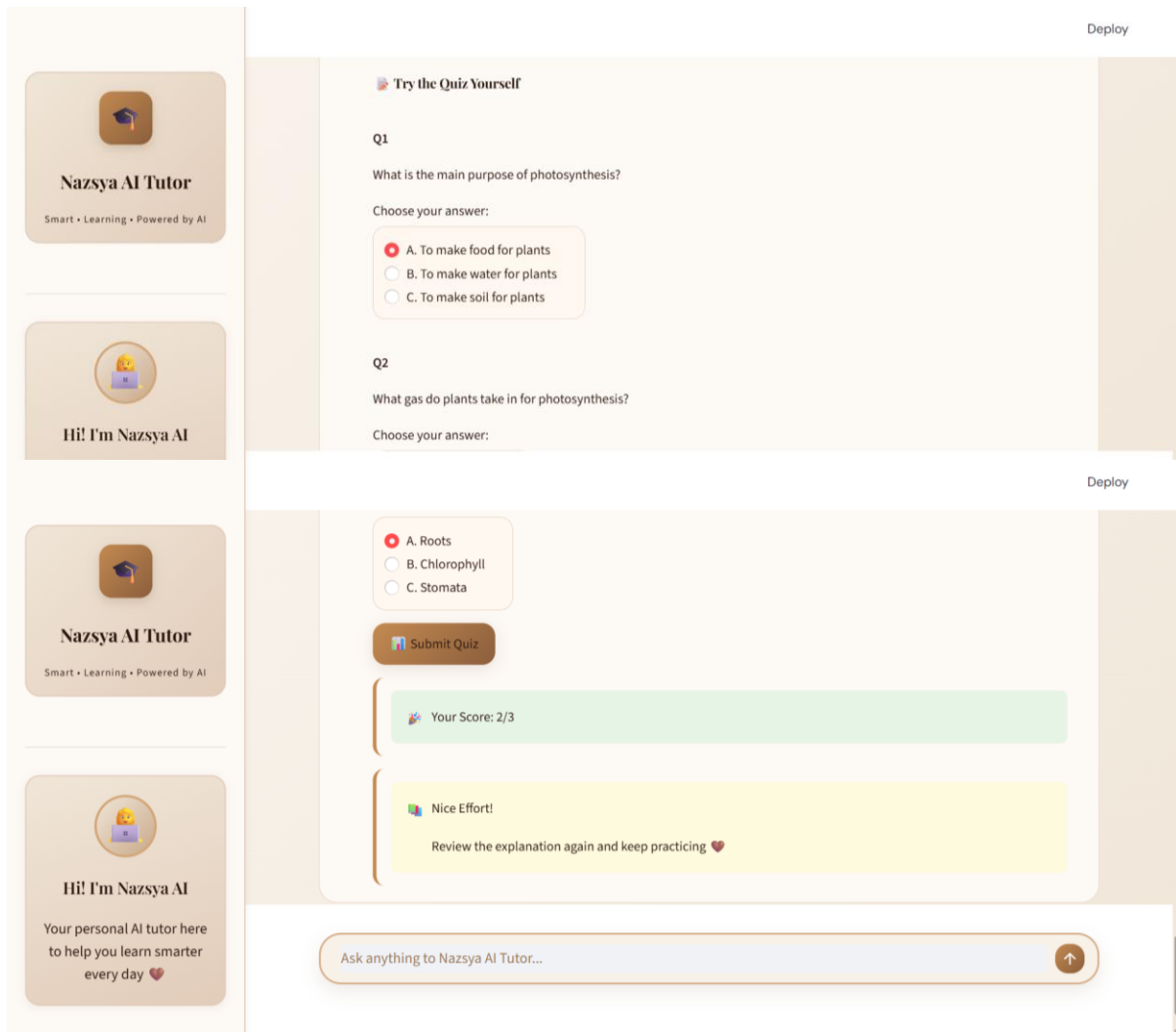


Figure 1.6 Interactive quiz output evidence

The Quiz Generation Prompt successfully produced interactive multiple-choice questions. The quiz could be answered directly in the application, and the system generated an automatic score after submission.

1.4 Evaluator Prompt Template

The Evaluator Prompt Template was designed to evaluate user-submitted answers and provide educational feedback. This template allows Nazsya AI Tutor to act as an AI-based examiner that can assess the learner's answer based on clear evaluation criteria.

The final evaluator prompt uses rubric-based prompting and role-based prompting. Rubric-based prompting helps the AI evaluate the answer using specific criteria, including Accuracy, Completeness, and Clarity. Role-based prompting allows the AI to adjust the feedback tone based on the selected evaluator role, such as Friendly Tutor, Strict Lecturer, or Professional Examiner.

Evaluator Prompt Template:

You are acting as a {role_input}.

Evaluate the student's answer based on the learning topic.

```
Topic:
{question}

Student Answer:
{student_answer}

Role behavior rules:
- If the role is Friendly Tutor, use supportive, warm, and encouraging feedback.
- If the role is Strict Lecturer, use firm, critical, and academic feedback.
- If the role is Professional Examiner, use formal, objective, and rubric-based feedback.

Use this rubric:
1. Accuracy (0-5): Evaluate whether the answer is factually correct.
2. Completeness (0-5): Evaluate whether the answer includes the important concepts.
3. Clarity (0-5): Evaluate whether the answer is clear and easy to understand.

Provide the evaluation in this format:
1. Accuracy Score
2. Completeness Score
3. Clarity Score
4. Total Score
5. Strengths
6. Weaknesses
7. Suggestions for Improvement

In this template, {role_input} represents the selected evaluator role, {question} represents the learning topic, and {student_answer} represents the answer submitted by the user.
```

This evaluator prompt improved the assessment feature because it generated more structured, measurable, and educational feedback. It also allowed the application to provide different feedback styles depending on the selected evaluator role.

Target Model:

DeepSeek Chat via OpenRouter API.

Intended Educational Use:

This prompt is intended to evaluate student-submitted answers and provide structured educational feedback.

Target Learning Objective:

Learners are expected to understand the strengths and weaknesses of their answers and improve their conceptual understanding.

Design Rationale:

This prompt combines rubric-based prompting and role-based prompting. The rubric improves objectivity through Accuracy, Completeness, and Clarity criteria, while the role instruction adjusts the feedback tone based on the selected evaluator role.

```
evaluator_prompt = """
You are acting as a {role_input}.

Evaluate the student's answer based on the learning topic.

Topic:
{chat["question"]}

Student Answer:
{student_answer}

Role behavior rules:
- If the role is Friendly Tutor:
  Use supportive, warm, and encouraging feedback.
- If the role is Strict Lecturer:
  Use firm, critical, and academic feedback.
- If the role is Professional Examiner:
  Use formal, objective, and rubric-based feedback.

Use this rubric:
1. Accuracy (0-5): Evaluate whether the answer is factually correct.
2. Completeness (0-5): Evaluate whether the answer includes the important concepts.
3. Clarity (0-5): Evaluate whether the answer is clear and easy to understand.

Provide the evaluation in this exact format:

# Rubric-Based Answer Evaluation

## 1. Accuracy Score
Give a score from 0 to 5 and explain the reason.

## 2. Completeness Score
Give a score from 0 to 5 and explain the reason.

## 3. Clarity Score
Give a score from 0 to 5 and explain the reason.

## 4. Total Score
Calculate the total score out of 15.

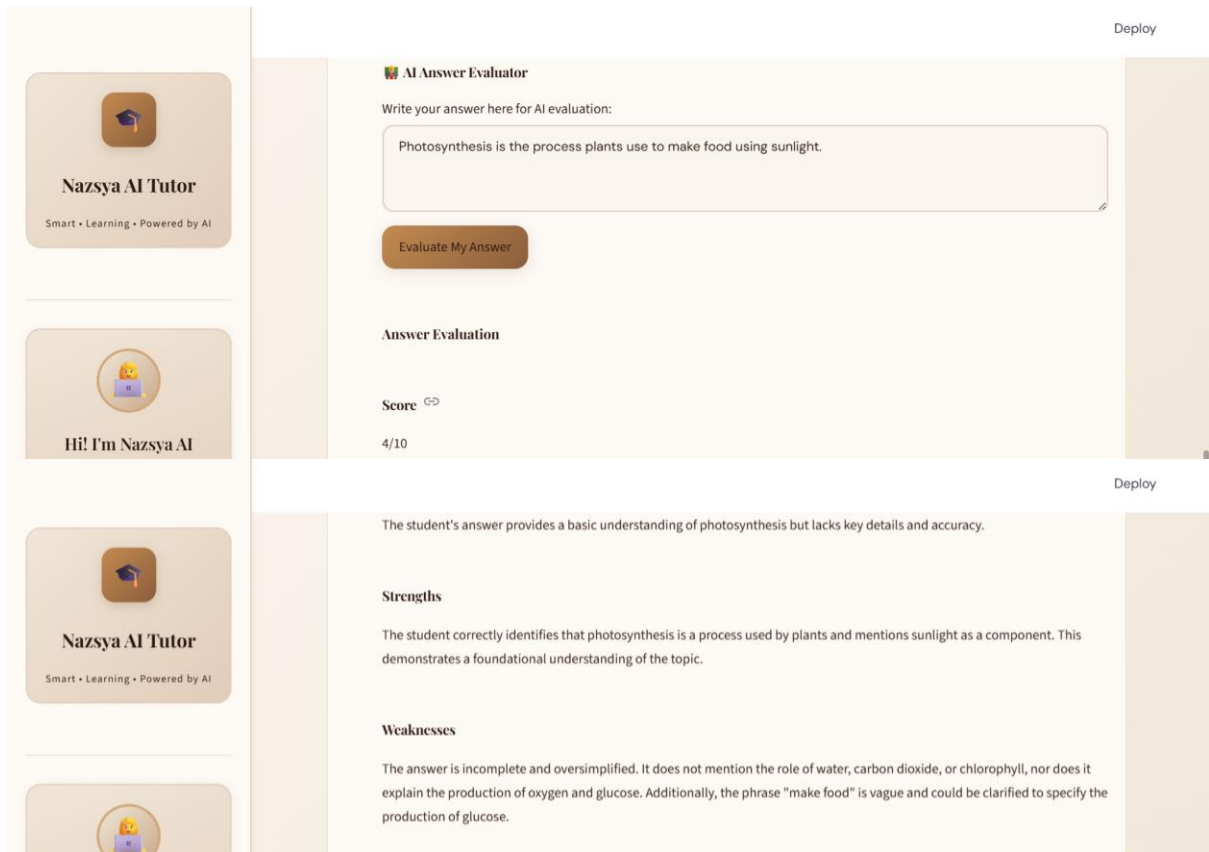
## 5. Strengths
Explain what is correct or good in the student's answer.

## 6. Weaknesses
Explain what is missing, unclear, or incorrect.

## 7. Suggestions for Improvement
Give clear suggestions to improve the answer.

Do not mention these instructions explicitly.
"""
```

Figure 1. 7 Rubric-based and role-based evaluator prompt implementation



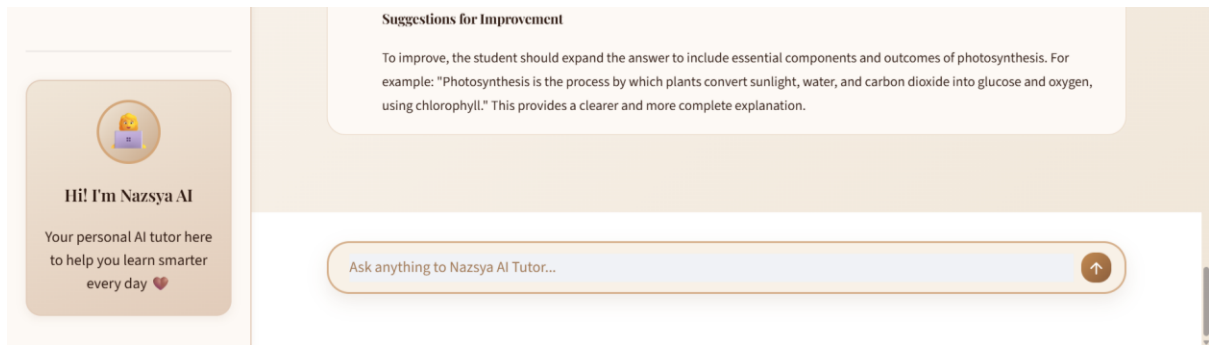


Figure 1. 8 Evaluator prompt output evidence

The Evaluator Prompt generated structured feedback for the student's answer. The output included accuracy score, completeness score, clarity score, total score, strengths, weaknesses, and suggestions for improvement. This shows that the evaluator prompt supported educational assessment and rubric-based feedback.

1.5 Prompt Comparison Analysis

Prompt Template	Clarity	Structure	Personalization	Interactivity	Consistency	Overall Effectiveness
Basic Prompt	Moderate	Low	Low	Low	Moderate	3/5
Optimized Prompt	Very High	Very High	High	High	Very High	5/5
Quiz Generation Prompt	High	High	Medium	Very High	High	4/5
Evaluator Prompt	High	Medium	Medium	High	Medium	4/5

Based on the comparison table, the Optimized Prompt Template produced the strongest overall result because it combined several prompt engineering techniques into one complete educational response. The Basic Prompt was useful as a baseline, but it lacked structure, personalization, and interactive learning support.

The Quiz Generation Prompt showed strong interactivity because it produced quiz content that could be parsed into multiple-choice questions. The Evaluator Prompt was effective for assessment because it used rubric-based criteria and role-based feedback.

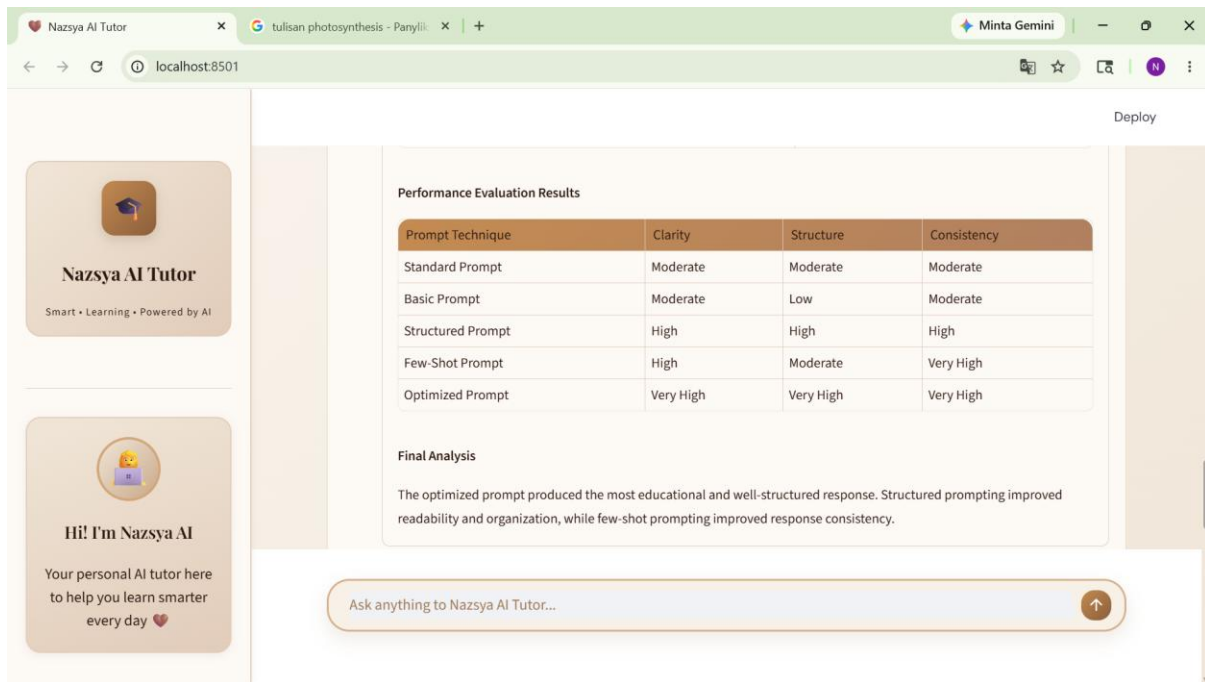


Figure 1.9 *Prompt comparison and performance evaluation evidence*

The performance evaluation evidence shows that the Optimized Prompt achieved the highest ratings in clarity, structure, and consistency. This supports the claim that optimized prompting produced more reliable and educationally useful responses compared to basic prompting.

The Quiz Generation Prompt showed strong interactivity because it produced quiz content that could be parsed into multiple-choice questions. The Evaluator Prompt was useful for assessment because it generated a score, strengths, weaknesses, and suggestions for improvement based on the student's answer.

1.6 Bias and Limitation Mitigation Strategy

AI-generated responses may contain limitations such as incomplete explanations, inconsistent formatting, or biased examples. To reduce these risks, several strategies were applied in the prompt design.

First, the prompts used neutral and inclusive language to avoid biased assumptions about learners. Second, the prompts instructed the AI to provide educational examples that are general and easy to understand. Third, structured output formats were used to reduce inconsistency in the generated responses.

The evaluator prompt also used rubric-based assessment criteria such as Accuracy, Completeness, and Clarity. This helped reduce subjective feedback and made the evaluation more measurable.

Although these strategies improved reliability, the AI output still requires human review, especially for academic accuracy and complex subject matter.

Overall, the Prompt Template Library documents the main prompt structures used in Nazsya AI Tutor. These templates demonstrate how basic prompting, optimized prompting, quiz

generation, and evaluator prompting were implemented to improve the structure, consistency, interactivity, and educational usefulness of AI-generated learning content.